1 Polar Coordinates and Complex Numbers

Central question. What happens when we describe a point by its distance and direction instead of its horizontal and vertical coordinates?

1.1 Two descriptions of the same point

Cartesian coordinates specify (x, y). Polar coordinates specify (r, θ), where r is the directed radial distance and θ is an angle from the positive horizontal axis:

$$\boxed{x = r\cos\theta, \qquad y = r\sin\theta.}$$

For $r \geq 0$, the distance is $r = \sqrt{x^2 + y^2}$. The angle must be chosen using the signs of both coordinates, not just $\arctan(y/x)$.

Example 1.1. The polar point $(4, 5\pi/6)$ corresponds to $(x, y) = (-2\sqrt{3}, 2)$. Its distance from the origin is 4, and it lies in Quadrant II.

Try it. Convert $(r, \theta) = (6, -\pi/3)$ to Cartesian coordinates. Then describe its direction using an angle in $[0, 2\pi)$.

1.2 Polar coordinates are not unique

Angles that differ by 2π represent the same direction, so (r, θ) and $(r, \theta + 2\pi k)$ describe the same point. A negative radius reverses the ray:

$$(r, \theta) = (-r, \theta + \pi).$$

For example, $(2, \pi/3)$, $(2, 7\pi/3)$, and $(-2, 4\pi/3)$ are three descriptions of the same point. At the origin $r = 0$, every angle describes the same location.

1.3 Complex numbers as plane vectors

Introduce i satisfying $i^2 = -1$. A complex number $z = a + bi$ corresponds to the point (a, b) in the complex plane. Its modulus and argument are

$$|z| = \sqrt{a^2 + b^2}, \qquad \arg z = \theta \quad \text{when } z = |z|(\cos\theta + i\sin\theta).$$

The argument of $z = 0$ is undefined. As with polar angles, nonzero arguments are only unique after choosing a branch, such as $(-\pi, \pi]$.

Example 1.2. For $z = -\sqrt{3} + i$, the modulus is 2 and one argument is $5\pi/6$. Thus

$$z = 2\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right).$$

1.4 Addition and multiplication mean different geometries

Complex addition follows the vector rule:

$$(a + bi) + (c + di) = (a + c) + (b + d)i.$$

Complex multiplication uses $i^2 = -1$:

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

If $z = r(\cos\alpha + i\sin\alpha)$ and $w = s(\cos\beta + i\sin\beta)$, the angle-addition formulas give

$$\boxed{zw = rs(\cos(\alpha+\beta) + i\sin(\alpha+\beta)).}$$

Therefore multiplying complex numbers multiplies distances from the origin and *adds angles*. A unit complex number acts as a rotation. This unifies the unit circle from Module 1 with vector geometry from Module 6.

Example 1.3. Multiplying $2+i$ by i gives $i(2+i) = -1+2i$. The vector $(2, 1)$ rotates counterclockwise by $90°$ to $(-1, 2)$, without changing its length.

1.5 De Moivre's theorem and roots

For integer n, repeated multiplication gives

$$\boxed{[r(\cos\theta + i\sin\theta)]^n = r^n(\cos n\theta + i\sin n\theta)}$$

whenever the expression makes sense; for negative n, require $r > 0$. For example, $(\cos(\pi/6) + i\sin(\pi/6))^3 = i$. To solve $z^n = R(\cos\phi + i\sin\phi)$ with $R > 0$, take

$$z_k = R^{1/n}\left(\cos\frac{\phi+2\pi k}{n} + i\sin\frac{\phi+2\pi k}{n}\right), \qquad k = 0, \ldots, n-1.$$

These n distinct roots are equally spaced around a circle.

Example 1.4. The four fourth roots of 1 are $1, i, -1, -i$. They are separated by a quarter-turn.

1.6 Polar graphs: a different way to draw a curve

In a polar equation such as $r = 2\cos\theta$, the radius changes with the angle. Multiply by r: $r^2 = 2r\cos\theta$, or $x^2 + y^2 = 2x$. Completing the square yields

$$(x-1)^2 + y^2 = 1.$$

Thus a simple polar equation can represent an ordinary circle displaced from the origin.

1.7 Exercises

1. Convert $(5, 2\pi/3)$ to Cartesian coordinates.
2. Give two distinct polar coordinate representations of $(\sqrt{2}, \sqrt{2})$.
3. Write $-1 - \sqrt{3}i$ in polar form with argument in $[0, 2\pi)$.
4. Calculate $(2 + i)(1 - 3i)$ in rectangular form.
5. Interpret multiplication by $-i$ as a plane transformation.
6. Evaluate $(\cos(\pi/4) + i\sin(\pi/4))^4$.
7. Find all three cube roots of 1 and plot them on the unit circle.
8. Convert the polar curve $r = 4\sin\theta$ into a Cartesian equation and identify the curve.